Derivation of Expected Accidental Coincidence Rates

Muon Shower Detector Array — Technical Note

Muon Technical Report v2, §6 (Expanded Derivation)

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1 Overview and Physical Motivation

An *accidental coincidence* is a near-simultaneous signal in multiple detectors that arises purely from independent, uncorrelated background events rather than from a genuine cosmic-ray air shower. Quantifying the expected rate of accidentals is essential: any statistically significant *excess* of observed coincidences above this background is evidence for real shower events.

The detector array consists of $N = 8$ scintillator panels (*Nano detectors*), each generating a time-stamped pulse whenever a charged particle passes through. After hardware calibration and a synchronisation veto to remove periodic artefacts, each detector independently fires at a roughly constant mean rate. We ask:

> *Given independent Poisson processes, how many times do we expect k specified detectors all to fire within a time window of half-width τ purely by chance, over an effective run of duration T_{eff}?*

The derivation proceeds in three stages:

1. Establish the Poisson framework and define the coincidence window.
2. Derive the two-fold ($k = 2$) accidental rate exactly.
3. Generalise to an arbitrary k-fold coincidence and sum over all detector subsets.

2 Definitions and Assumptions

Definition 1 (Detector rates and run duration)**.** Let detector i ($i = 1, \ldots, N$, $N = 8$) fire according to an independent, stationary Poisson process with measured rate r_i (events s^{-1}). The effective run duration after dead-time and veto corrections is T_{eff} (s).

Definition 2 (Coincidence window)**.** A *coincidence window* of half-width τ is the symmetric interval $[t-\tau,\ t+\tau]$ of total width 2τ centred on a reference time t. Two events are coincident if they fall within the same window.

Assumption 1 (Stationarity)**.** Each r_i is constant over the run. This is verified after calibration and the sync veto.

Assumption 2 (Independence between detectors)**.** The firing of detector j is statistically independent of the firing of detector $i \neq j$. A significant excess of observed coincidences over the accidental prediction derived below is therefore evidence of *correlated* events (genuine showers).

Assumption 3 (Rare-event / thin-window approximation)**.** For the window widths and rates of interest,

$$r_i \cdot 2\tau \ \ll\ 1. \tag{1}$$

Concretely, at $r \approx 1$ Hz and $\tau = 0.5$ ms this factor is $r \cdot 2\tau = 10^{-3}$, accurate to 0.1 %. Under this condition the probability of *two or more* events from the same detector falling inside a single window is $\mathcal{O}[(r\tau)^2]$ and is negligible.

3 Poisson Background: Key Probability Results

Before deriving the coincidence rates we collect the Poisson results that underpin the calculation.

3.1 Number of events in a fixed interval

A Poisson process with rate r generates a number of events X in a time interval of length Δt distributed as

$$P(X = n) = \frac{(r\,\Delta t)^n\, e^{-r\,\Delta t}}{n!}, \qquad n = 0, 1, 2, \ldots \tag{2}$$

with mean $\langle X \rangle = r\,\Delta t$ and variance $\mathrm{Var}(X) = r\,\Delta t$.

3.2 Probability of at least one event

The probability that at least one event falls in Δt is

$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-r\,\Delta t}. \tag{3}$$

Under the rare-event approximation (1) (i.e. $r\,\Delta t \ll 1$), the Taylor expansion of (3) gives

$$\begin{aligned} P(X \geq 1) &= 1 - e^{-r\,\Delta t} \\ &= 1 - \left[1 - r\,\Delta t + \tfrac{(r\,\Delta t)^2}{2} - \cdots\right] \\ &= r\,\Delta t - \tfrac{(r\,\Delta t)^2}{2} + \cdots \\ &\approx r\,\Delta t. \end{aligned} \tag{4}$$

The discarded terms are $\mathcal{O}[(r\,\Delta t)^2]$, which by Assumption 3 are $\lesssim 10^{-6}$ for $r \sim 1$ Hz and $\tau \leq 10$ ms.

Physical interpretation. In the rare-event regime the window is so narrow that the probability of getting *any* hit from a single detector equals simply the mean number of hits expected in the window. One can think of $r_j \cdot 2\tau$ as the fractional "dead fraction" of the run that detector j's activity fills.

4 Two-Fold Accidental Coincidence Rate

4.1 Setup: trigger-and-follower framework

Consider a specific ordered pair of detectors (i, j). We use a *trigger–follower* argument:

1. **Trigger:** Detector i fires at time t. This defines a coincidence window $W = [t-\tau,\, t+\tau]$ of width 2τ.
2. **Follower:** We ask whether detector j (independent of i by Assumption 2) also fires at least once inside W.

The key insight is that, conditioned on i having fired, the subsequent behaviour of j inside W is simply a Poisson process with rate r_j acting over a fixed time interval of length 2τ.

4.2 Probability that the follower fires in the window

From (4), the probability that detector j fires at least once in the window W is

$$P(j \text{ fires in } W \mid i \text{ triggered}) = 1 - e^{-r_j \cdot 2\tau} \;\approx\; r_j \cdot 2\tau, \tag{5}$$

where the approximation uses (4) and Assumption 3.

4.3 Expected number of accidental two-fold coincidences

Over the entire run, detector i produces a total of

$$N_i = r_i \, T_{\text{eff}} \tag{6}$$

trigger events (in expectation). Each trigger independently opens a window, and each window has probability (5) of capturing a hit from j. These trials are:

- *Identical:* the same probability applies to every window (stationarity, Assumption 1).
- *Independent:* non-overlapping triggers are by definition independent; even for closely spaced triggers the independence follows from the memoryless property of Poisson processes.

Therefore the expected number of accidental coincidences between the specific pair (i, j) is

$$\begin{aligned} E_{2,ij} &= \underbrace{r_i \, T_{\text{eff}}}_{\text{no. of triggers}} \times \underbrace{r_j \cdot 2\tau}_{\text{prob. follower hits window}} \\ &= r_i \, r_j \, (2\tau) \, T_{\text{eff}}. \end{aligned} \tag{7}$$

Result 1 (Two-fold accidental rate, Eq. (6.1))**.**

$$\boxed{E_{2,ij} = r_i \, r_j \, (2\tau) \, T_{\text{eff}}} \tag{6.1}$$

4.4 Symmetry check

Expression (7) is *symmetric* in i and j: swapping which detector is the trigger gives

$$E_{2,ji} = r_j \, r_i \, (2\tau) \, T_{\text{eff}} = E_{2,ij}.$$

This is physically required: the expected number of accidental coincidences between a pair of detectors cannot depend on our bookkeeping choice of which one we call the "trigger."

4.5 Double-counting note

When we later sum over all *pairs*, the combination $\{i, j\}$ is counted once (as an unordered subset). We do **not** add both $E_{2,ij}$ and $E_{2,ji}$; the trigger is a mathematical device, not a physical distinction.

5 k-Fold Accidental Coincidence Rate for a Specific Subset

5.1 Generalising the trigger-follower argument

Let $S = \{i_1, i_2, \ldots, i_k\}$ be a specific ordered set of k detectors. Designate i_1 as the trigger.

Step 1: Trigger events. Detector i_1 fires $r_{i_1} T_{\text{eff}}$ times (in expectation) over the run. Each firing at time t opens a window $W = [t - \tau, t + \tau]$.

Step 2: Each follower fires independently. For each follower detector i_m ($m = 2, \ldots, k$), the probability that it fires at least once inside W is (from (5))

$$P(i_m \text{ fires in } W \mid i_1 \text{ triggered}) \approx r_{i_m} \cdot 2\tau. \tag{8}$$

Step 3: All $k-1$ followers fire simultaneously. By Assumption 2 (independence between detectors), the events in (8) are independent for different m. The probability that *all* $k-1$ followers fire inside W is therefore the product of the individual probabilities:

$$P(\text{all followers fire in } W \mid i_1 \text{ triggered}) = \prod_{m=2}^{k} r_{i_m} \cdot 2\tau = (2\tau)^{k-1} \prod_{m=2}^{k} r_{i_m}. \tag{9}$$

The $(2\tau)^{k-1}$ factor arises because there are $k-1$ followers each contributing one power of 2τ.

5.2 Combining trigger count and follower probability

Multiplying the number of triggers (6) by the joint follower probability (9):

$$\begin{aligned} E_{k,S} &= r_{i_1} T_{\text{eff}} \ \times \ (2\tau)^{k-1} \prod_{m=2}^{k} r_{i_m} \\ &= T_{\text{eff}} \, (2\tau)^{k-1} \ \underbrace{r_{i_1} \prod_{m=2}^{k} r_{i_m}}_{=\prod_{m=1}^{k} r_{i_m}} \\ &= T_{\text{eff}} \, (2\tau)^{k-1} \prod_{m=1}^{k} r_{i_m}. \end{aligned} \tag{10}$$

Result 2 (k-fold accidental rate for a specific k-tuple, Eq. (6.2))**.**

$$\boxed{E_{k,S} = T_{\text{eff}} \, (2\tau)^{k-1} \prod_{i \in S} r_i} \tag{6.2}$$

5.3 Why the choice of trigger drops out

The final expression (10) contains the product of *all* k rates $r_{i_1} \cdots r_{i_k}$, regardless of which detector was labelled the trigger. The trigger contributed one factor of r_{i_1} to the count $N_{i_1} = r_{i_1} T_{\text{eff}}$, while the $k-1$ followers each contributed one factor of $r_{i_m} \cdot 2\tau$. The product simply collects all k rates symmetrically. In other words, had we chosen i_2 as the trigger instead, the derivation would have given

$$r_{i_2} T_{\text{eff}} \times (2\tau)^{k-1} \prod_{\substack{m=1 \\ m \neq 2}}^{k} r_{i_m} = T_{\text{eff}} \, (2\tau)^{k-1} \prod_{m=1}^{k} r_{i_m},$$

which is identical. The result is fully symmetric in all k detectors.

5.4 Dimensional analysis

Quantity	Units	Exponent in (10)
T_{eff}	s	1
$(2\tau)^{k-1}$	s^{k-1}	$k-1$
$\prod r_{i_m}$	$(\text{s}^{-1})^k$	k
Product	$\text{s}^{1+(k-1)-k} = \text{s}^0 =$ dimensionless	

The result is a dimensionless *count*, as required.

6 Total Expected k-Fold Accidentals (All Subsets)

6.1 Summing over all k-element subsets

The array contains $N = 8$ detectors. The number of distinct k-element subsets (unordered combinations) is $\binom{N}{k} = \binom{8}{k}$. The total expected number of accidental k-fold coincidences is the sum of (10) over all such subsets:

$$E_k = \sum_{\substack{S \subseteq \{1,\ldots,8\} \\ |S|=k}} T_{\text{eff}} \, (2\tau)^{k-1} \prod_{i \in S} r_i. \tag{11}$$

Since T_{eff} and $(2\tau)^{k-1}$ do not depend on the subset S, they factor out of the sum:

$$\boxed{E_k = T_{\text{eff}} \, (2\tau)^{k-1} \sum_{\substack{S \subseteq \{1,\ldots,8\} \\ |S|=k}} \prod_{i \in S} r_i.} \tag{6.3}$$

Result 3 (Total k-fold accidental rate, Eq. (6.3))**.** Equation (6.3) is exact within the Poisson framework and the rare-event approximation. It is implemented numerically in `expected_accidentals()` using the elementary symmetric polynomial $e_k(r_1, \ldots, r_8)$ to enumerate all $\binom{8}{k}$ subsets efficiently.

6.2 Connection to elementary symmetric polynomials

The sum $\sum_{|S|=k} \prod_{i \in S} r_i$ is precisely the k-th *elementary symmetric polynomial* in the rates:

$$e_k(r_1, \ldots, r_N) = \sum_{1 \le i_1 < i_2 < \cdots < i_k \le N} r_{i_1} r_{i_2} \cdots r_{i_k}.$$

For instance,

$$e_2(r_1, \ldots, r_8) = \sum_{i<j} r_i r_j, \qquad e_3(r_1, \ldots, r_8) = \sum_{i<j<\ell} r_i r_j r_\ell, \quad \ldots$$

so that $E_k = T_{\text{eff}} (2\tau)^{k-1} \, e_k(r_1, \ldots, r_8)$.

6.3 Scaling with fold order

A key insight is how E_k falls as k increases. For simplicity suppose all rates are equal: $r_i = r$. Then $e_k(r, \ldots, r) = \binom{8}{k} r^k$, so

$$E_k = T_{\text{eff}} \, (2\tau)^{k-1} \binom{8}{k} r^k = T_{\text{eff}} \, r \binom{8}{k} (r \cdot 2\tau)^{k-1}.$$

The ratio of consecutive fold orders is

$$\frac{E_{k+1}}{E_k} = \frac{\binom{8}{k+1}}{\binom{8}{k}} \cdot r \cdot 2\tau = \frac{8-k}{k+1} \cdot r \cdot 2\tau. \tag{12}$$

At $r = 1$ Hz and $2\tau = 1$ ms, the factor $r \cdot 2\tau = 10^{-3}$, so each additional fold suppresses E_k by roughly three orders of magnitude (multiplied by a combinatorial factor of order unity). This is why a single observed 5-fold event, when $E_5 \sim 10^{-3}$ at the 1 ms window, constitutes an enormous statistical excess.

7 Numerical Results for the February 27 Dataset

The following table reproduces the expected accidentals from the Technical Report (§6.3) for the February 27, 2026 run ($T_{\text{eff}} \approx 3{,}576$ s, rates $r_i \in [0.81, 1.25]$ Hz):

To illustrate the scaling law (12) concretely, at $2\tau = 1$ ms:

$$\frac{E_3}{E_2} \approx \frac{0.80}{202} \approx 4 \times 10^{-3} \approx \frac{7}{2} \cdot r_{\text{avg}} \cdot 2\tau = \frac{7}{2} \cdot 1\,\text{Hz} \cdot 10^{-3}\,\text{s},$$

consistent with (12) at $k = 2$.

8 Validity Conditions

The derivation above rests on three conditions that must hold in practice.

Table 1: Expected accidental coincidence counts E_k from Eq. (6.3) for selected coincidence windows and fold orders. Values are rounded to two significant figures.

2τ	E_2	E_3	E_4	E_5	E_6
1 ms	202	0.80	$< 10^{-2}$	$< 10^{-3}$	$< 10^{-4}$
2 ms	405	3.2	$< 10^{-2}$	$< 10^{-3}$	$< 10^{-4}$
5 ms	1 012	20.3	0.3	$< 10^{-2}$	$< 10^{-3}$
10 ms	2 024	81.1	2.0	0.02	$< 10^{-3}$

1. Poisson statistics (stationarity and independence within a detector). Each detector must fire at a constant mean rate with no clustering or periodicity in its own pulse stream. The synchronisation veto removes hardware-generated periodic artefacts, making the post-veto event stream consistent with Poisson statistics. Stationarity is verified by checking that the measured rate r_i is stable across short sub-runs.

2. Rare-event / thin-window approximation. Equation (1) must hold: $r_i \cdot 2\tau \ll 1$. For the rates and windows in Table 1:

2τ	$r_{\max} \cdot 2\tau$	Fractional error in E_k
1 ms	1.25×10^{-3}	$\sim 0.1\,\%$
10 ms	1.25×10^{-2}	$\sim 1\,\%$

At all windows used, the approximation error is well within the statistical uncertainty of the measurements.

3. Independence between detectors. Equation (9) multiplies probabilities for different detectors using the product rule, which requires independence (Assumption 2). Physically this holds if the detectors do not share a noise source and if cosmic-ray showers — which would create correlated hits — are rare. The entire point of the accidental calculation is to define a baseline against which genuine shower correlations are detected as a statistically significant excess.

9 Summary of Results

Result 4 (Two-fold accidentals between a specific pair (i, j))**.**

$$E_{2,ij} = r_i\, r_j\, (2\tau)\, T_{\text{eff}}. \tag{6.1}$$

Result 5 (k-fold accidentals for a specific k-tuple S)**.**

$$E_{k,S} = T_{\text{eff}}\, (2\tau)^{k-1} \prod_{i \in S} r_i. \tag{6.2}$$

Result 6 (Total k-fold accidentals summed over all subsets)**.**

$$E_k = T_{\text{eff}}\,(2\tau)^{k-1} \sum_{\substack{S \subseteq \{1,\ldots,8\} \\ |S|=k}} \prod_{i \in S} r_i = T_{\text{eff}}\,(2\tau)^{k-1}\, e_k(r_1, \ldots, r_8). \tag{6.3}$$

The steep suppression with k — approximately a factor of $r \cdot 2\tau \approx 10^{-3}$ per additional fold at 1 ms windows — means that even a single observed high-fold coincidence event constitutes an overwhelming statistical excess above this accidental background.